

CVN-69 Milestone 6 — Deck Vehicle Drawings

Equipment dimensions and released manufacturing envelopes

Family	Public full-scale envelope	Released 1:700 FDM envelope	Accuracy boundary
STT49 A/S32A-49 Shipboard Tow Tractor	3528 × 1859 × 1020 mm	5.040 × 3.400 × 2.180 mm	manufacturer-dimension-derived; official-photo-informed silhouette; deliberately enlarged for FDM printability
P25A A/S32P-25A Shipboard Firefighting Vehicle	4826 × 1778 × 1626 mm	6.897 × 3.250 × 2.860 mm	official-dimension-derived; official-photo-informed silhouette; deliberately enlarged for FDM printability
MSU200 MSU-200NAV Air Start Service Cart	1735 × 930 × 695 mm	3.600 × 2.550 × 1.900 mm	official-dimension-derived module on photo-informed towable frame; deliberately enlarged for FDM printability
TOWBAR Carrier Aircraft Tow Bar	3500 × 450 × 300 mm	4.725 × 2.200 × 0.800 mm	official-photo-derived; visually approximated; deliberately enlarged for FDM printability
LADDER Portable Aircraft Maintenance Ladder	1800 × 900 × 2200 mm	3.600 × 0.600 × 3.850 mm	deployment-era official-photo-derived; visually approximated; deliberately enlarged for FDM printability
CHOCK Aircraft Wheel Chock Group	850 × 500 × 300 mm	3.100 × 1.100 × 0.700 mm	official-photo-derived generic aviation safety form; visually approximated; deliberately enlarged for FDM printability
EXT Twin Portable Flight-Deck Extinguisher Cart	1100 × 650 × 1200 mm	2.600 × 2.300 × 2.200 mm	official-equipment-description-derived; visually approximated; deliberately enlarged for FDM printability

Manufacturing sections and part breakdown

Feature	Released minimum	Use
structural wall / body section	0.80 / 1.00 mm	solid static bodies
wheel	1.00 mm diameter × 0.70 mm width	black glue-on insert
axle / tow bar	0.80 mm	bed-connected support / yellow gear
handle / post	0.70 mm	static FDM enlargement
ladder / hose	0.60 mm	flat-backed ladder / reel
raised detail	0.50 × 0.30 mm	minimum visible relief
color insert	0.60 mm	window, wheel, turret/reel modules

Print orientation and sprues

- Bodies and inserts use engineered z=0 faces. Wheel inserts use bed-connected underbody rails and four distinct end wheels.
- The maintenance ladder prints flat on its back. The tow-bar and chock sprues use 0.80 mm family-only gates on non-visible surfaces.
- All multi-part interfaces are glue-only; the first-article coupon releases 0.20 mm clearance per side, independent of the frozen ship interface.

